ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

43rd HUST Annual Student Research Conference

A Collaborative Multi-agent Reinforcement Learning-Based Framework for Automated Related Work Summarization

ID: S.1.17

Students: Dang Hai Dang, Pham Bao Yen, Nguyen Quang Huy, Do Truong Giang

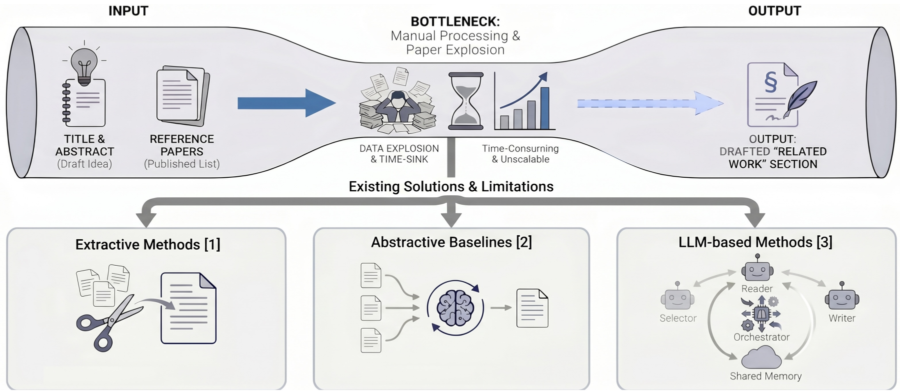
Supervisor: Prof. Huynh Thi Thanh Binh, Dr. Nguyen Thi Huong

Hanoi University of Science and Technology

May 15, 2026

ONE LOVE. ONE FUTURE.

Problem Statement



[1] Cong Duy Vu Hoang and Min-Yen Kan. "Towards automated related work summarization". In: *Coling 2010: Posters*. 2010, pp. 427–435

[2] Wen Xiao et al. "PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263

[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication

Solution

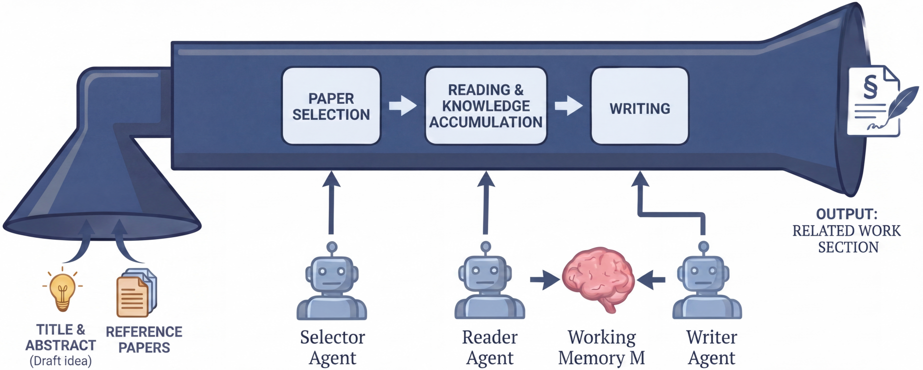


Figure 1: Prior work leveraging LLM-based method.

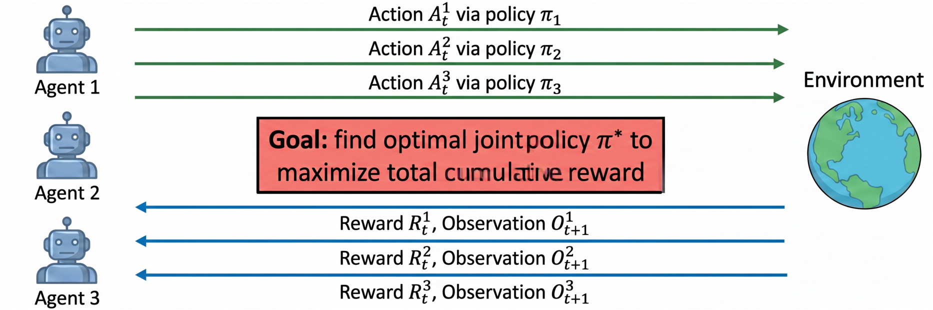


Figure 1: A Collaborative Multi-agent Reinforcement Learning-Based Framework (CREW).

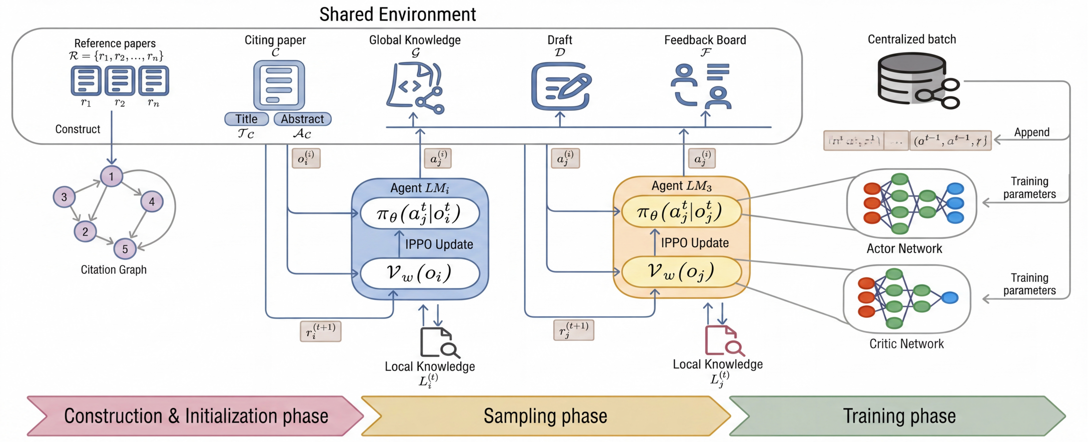


Figure 2: Proposed system architecture overview.

Experimental Setup and Results

Experimental Setup

- Environment: Python 3.10, NVIDIA A100 GPU (Train & Infer: Qwen-32B-Instruct).
- Dataset: OARelatedWork [4].
- Metrics: LLM-as-a-Judge (inc. Coverage, Logic, Relevance) [3].
- Baselines: PRIMERA [2], GPT-4o, Select-RW [3].

Results

Table 1. Effectiveness

Model	Cov.	Log.	Rel.	Avg.	Tok.
PRIMERA	2.48	2.71	3.02	2.64	186K
GPT-4o	3.18	4.08	3.96	3.36	2.1K
Select-RW	3.36	4.21	4.18	3.53	186K
CREW	**3.62**	**4.39**	**4.31**	**3.77**	**153K**

Table 2. Ablation Study

Config	Cov.	Log.	Rel.	Avg.	Tok.
w/o Draft	3.41 (-.21)	4.02 (-.37)	4.10 (-.21)	3.50 (-.27)	150K
w/o Crit.	3.35 (-.27)	4.08 (-.31)	4.02 (-.29)	3.42 (-.35)	143K
w/o BB.	3.29 (-.33)	3.95 (-.44)	3.98 (-.33)	3.35 (-.42)	160K
CREW	**3.62**	**4.39**	**4.31**	**3.77**	**153K**

[4] Martin Docekal, Martin Fajcik, and Pavel Smrz. "Oarelatedwork: A large-scale dataset of related work sections with full-texts from open access sources". In: *arXiv preprint arXiv:2405.01930* (2024)
[2] Wen Xiao et al. "PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263
[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication

Case Study

Target Paper: Robust Non-Rigid Registration of 2D and 3D Graphs

Abstract: We present a new approach to matching graphs in $\mathbb{R}^2$ or $\mathbb{R}^3$. Our method handles partial matches and non-linear deformations using Gaussian Processes, without requiring appearance features or initial alignment... [...]

Baseline: Select-RW [3]

Advanced techniques have emerged to overcome these limitations. One notable approach is the SoftPosit algorithm. Despite its effectiveness, the SoftPosit algorithm can still fail in the presence of clutter or occlusions **[3755653]**.

To address these limitations, a recent paper proposes an innovative solution by integrating a GMM to refine pose priors **[3755653]**. This method progressively refines the pose estimates by hypothesizing correspondences. By reducing matches and exploring the pose space more thoroughly, this approach enhances the robustness and efficiency **[3755653]**.

Ours: CREW

While [28637] and [27485] focus on geometric and appearance priors, other methodologies such as spectral methods and deep learning have also been explored. Spectral methods, which leverage the eigenvalues and eigenvectors of graph Laplacians, offer a powerful tool for graph matching by capturing the intrinsic structure. These methods are particularly useful for regular structures where dominant eigenvalues are present. However, they struggle with nonlinear deformations, which are better handled by geometric approaches in [28637].

[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication

Conclusions and Future Work

- **Conclusions**
  - **Collaborative Framework:** Proposed CREW, a MARL-based system for automated related work summarization.
  - **Empirical Success:** Improved coverage, relevance, and logic through dynamic agent coordination.
- **Future Work**
  - **System Expansion:** Integrate external database crawling and dynamic online document graphs.

THANK YOU FOR LISTENING!

References I

[1] Cong Duy Vu Hoang and Min-Yen Kan. "Towards automated related work summarization". In: *Coling 2010: Posters*. 2010, pp. 427–435.

[2] Wen Xiao et al. "PRIMERA: Pyramid-based Masked Sentence Pre-training for Multi-document Summarization". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2022, pp. 5245–5263.

[3] Xiaochuan Liu et al. "Select, Read, and Write: A Multi-Agent Framework of Full-Text-based Related Work Generation". In: *Findings of the Association for Computational Linguistics* (2025). Accepted for publication.

[4] Martin Docekal, Martin Fajcik, and Pavel Smrz. "Oarelatedwork: A large-scale dataset of related work sections with full-texts from open access sources". In: *arXiv preprint arXiv:2405.01930* (2024).